

Project Figures



Figure 1: Live Accelerometer data when stationary

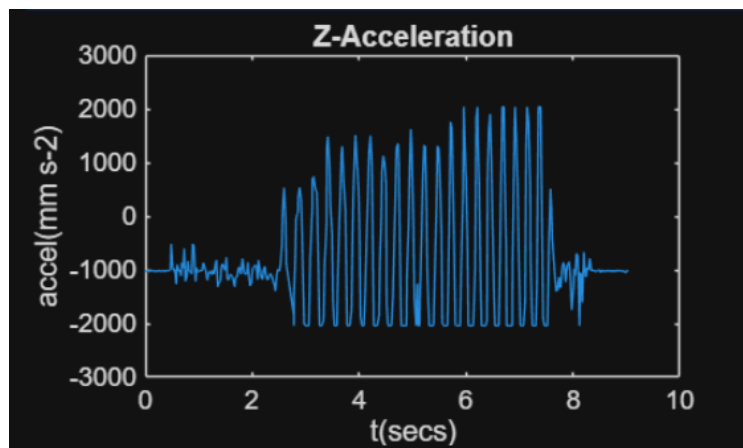


Figure 2: The Z-acceleration (which the microbit clips to a magnitude of 2g), which was applied to the microbit

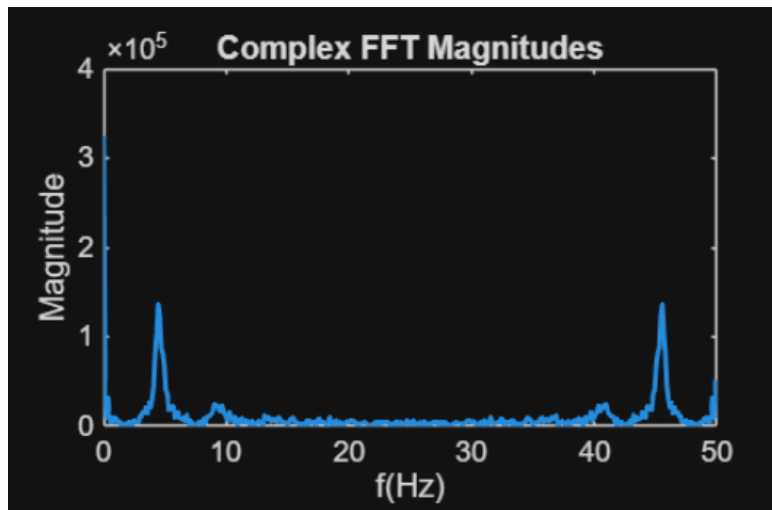


Figure 3: MATLAB's inbuilt FFT Spectrum Plot of Figure 2

One interesting thing to note is the symmetry of the plot. As the FFT is carried out using complex numbers, which cycle around every 2π , any frequency peaks also appear as peaks at $F_s - f_{\text{peak}}$, where F_s is the sampling frequency/rate.

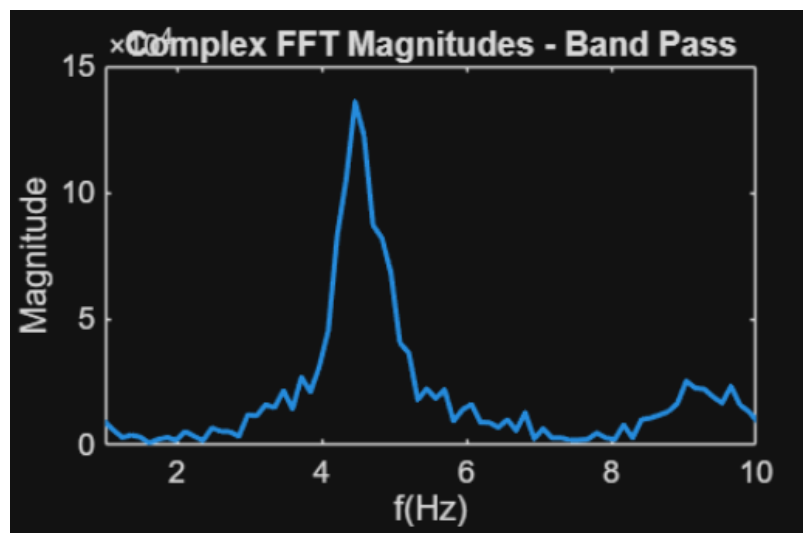


Figure 4: A zoomed-in spectrum plot from Figure 3

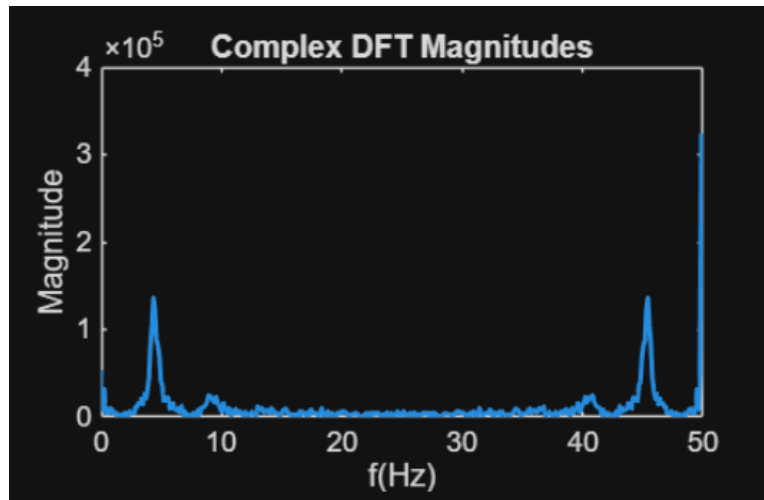


Figure 5: The manual DFT algorithm's spectrum plot, of the same data

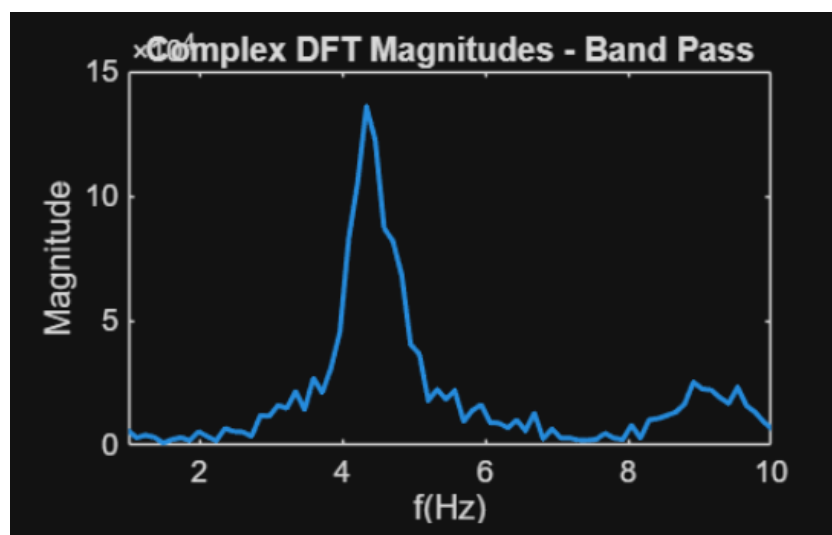


Figure 6: The same plot as Figure 5 but zoomed in, thus verifying that the two algorithms produce the same results.